

Appendix A. Intrinsic Derivation of the Proposed Dimensionality Reduction under the Affine-Invariant Riemannian Metric

Appendix A.1. Affine-Invariant Riemannian Metric on the SPD Manifold

Let $Sym^+(d)$ denote the manifold of $d \times d$ symmetric positive definite (SPD) matrices. This manifold admits a canonical affine-invariant Riemannian metric (AIRM), which has been widely used in statistical analysis of covariance matrices.

Under this metric, the geodesic distance between $p, q \in Sym^+(d)$ is defined as:

$$d_{\text{AIRM}}^2(p, q) = \left\| \log \left(p^{-\frac{1}{2}} q p^{-\frac{1}{2}} \right) \right\|_F^2. \quad (\text{A.1})$$

This distance is determined by the congruence transformation $p^{-\frac{1}{2}} q p^{-\frac{1}{2}}$, which characterizes the intrinsic geometric relationship between p and q under the affine-invariant structure. This transformation can be viewed as a relative representation of q with respect to the reference point p . Particularly, the interaction between two SPD matrices can be described through the congruence-invariant quantity $p_{ij} = p_i^{-\frac{1}{2}} p_j p_i^{-\frac{1}{2}}$. Such a representation has been employed in the analysis of dynamical brain functional connectivity[2].

Appendix A.2. Objective Function Derivation

Let $\{p_i\}_{i=1}^N \subset Sym^+(d_1)$ be a collection of SPD matrices, and let $\mathcal{S}_{d_1, d_2} = \{B \in \mathbb{R}^{d_1 \times d_2} : B^T B = I_{d_2}\}$ denote the Stiefel manifold. Based on the affine-invariant Riemannian structure introduced above, we consider the intrinsic geometry of $Sym^+(d_1)$ relative to the identity matrix as a common base point. In this case, the representation reduces to $d(I, p_i) = I^{-1/2} p_i I^{-1/2} = p_i$. Hence, the intrinsic structure relative to the identity is fully encoded by the matrices $\{p_i\}$ themselves.

To perform dimensionality reduction, we introduce an orthogonal projection matrix $B \in \mathcal{S}_{d_1, d_2}$ with $B^T B = I_{d_2}$ and define the low-dimensional representation $q_i = B^T p_i B \in Sym^+(d_2)$. In order to compare the projected matrices with the original ones within the same ambient space, we reconstruct them as $\hat{p}_i = B q_i B^T = B B^T p_i B B^T$. Following [2], the structural distortion is measured by the Frobenius discrepancy between the original and reconstructed matrices:

$$\begin{aligned}
E(B) &= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_i (\mathrm{d}_{\mathrm{Sym}^+(d_1)}(\mathbf{I}_{d_1}, p_i) - \mathrm{d}_{\mathrm{Sym}^+(d_1)}(\mathbf{I}_{d_1}, \hat{p}_i))^2 \\
&= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_i \left\| \mathbf{I}^{-\frac{1}{2}} p_i \mathbf{I}^{-\frac{1}{2}} - \mathbf{I}^{-\frac{1}{2}} \hat{p}_i \mathbf{I}^{-\frac{1}{2}} \right\|^2 \\
&= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_i \|p_i - B q_i B^T\|^2 \\
&= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_i \|p_i - B B^T p_i B B^T\|^2 \\
&= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_{i=1} \operatorname{tr} \left((p_i - B B^T p_i B B^T)^2 \right) \\
&= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_{i=1} \left[\operatorname{tr}(p_i^2) - \operatorname{tr}(B^T p_i B B^T p_i B) \right].
\end{aligned}$$

Note that the term $\sum_{i=1}^N \operatorname{tr}(p_i^2)$ does not depend on B and is therefore constant with respect to the optimization variable. Consequently, it does not affect the location of the minimizer. The optimization problem is thus equivalent to minimizing $-\sum_{i=1}^N \operatorname{tr}(B^T p_i B B^T p_i B)$ over $B \in \mathcal{S}_{d_1, d_2}$, which is in turn equivalent to maximizing $\sum_{i=1}^N \operatorname{tr}(B^T p_i B B^T p_i B)$ on the same manifold. Hence, the dimensionality reduction problem reduces to

$$\operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} - \sum_{i=1}^N \operatorname{tr}(B^T p_i B B^T p_i B) = \operatorname{argmax}_{B \in \mathcal{S}_{d_1, d_2}} \sum_{i=1}^N \operatorname{tr}(B^T p_i B B^T p_i B).$$

This formulation explicitly preserves the distances from each SPD matrix to the identity matrix, ensuring that the low-dimensional projections maintain the essential geometric structure of the original data.

Appendix A.3. Riemannian Gradient on the Stiefel Manifold

To optimize the proposed dimensionality reduction objective on the Stiefel manifold, we first derive the Euclidean gradient of $f(B) = \sum_{i=1}^N \operatorname{tr}((B^T p_i B)^2)$ for $B \in \mathcal{S}_{d_1, d_2}$. Letting $c_i = B^T p_i B$, the differential of c_i with respect to B is

$dc_i = (dB)^T p_i B + B^T p_i (dB)$, which yields the differential of $f(B)$:

$$\begin{aligned}
df(B) &= 2 \sum_{i=1}^N \text{tr}(c_i dc_i) \\
&= 2 \sum_{i=1}^N \text{tr}(c_i (dB)^T p_i B) + 2 \sum_{i=1}^N \text{tr}(c_i B^T p_i (dB)) \\
&= 2 \sum_{i=1}^N \text{tr}((p_i B c_i + p_i B c_i)^T dB) \\
&= 4 \sum_{i=1}^N \text{tr}((p_i B B^T p_i B)^T dB).
\end{aligned}$$

Hence, the Euclidean gradient is $\nabla f(B) = 4 \sum_{i=1}^N p_i B B^T p_i B$. To obtain the Riemannian gradient of $f(B)$ on the Stiefel manifold $\mathcal{S}_{d_1, d_2}$, we project the Euclidean gradient $\nabla f(B)$ onto the tangent space at B :

$$\text{grad } f(B) = \nabla f(B) - B \text{sym}(B^T \nabla f(B)),$$

where $\text{sym}(X) = \frac{1}{2}(X + X^T)$ denotes the symmetric part of a matrix. This projected gradient is then used in the Manopt toolbox to perform optimization on the manifold.

Appendix B. Iterative Estimation Algorithm for ITS RM

We introduce an iterative optimization strategy to solve the proposed model, jointly estimating the parametric components (p_t, V_t) and the non-parametric component $g_t(\cdot)$.

$$y_t^i = \text{Exp}(p_t, V_t x^i + g_t(z^i) + \varepsilon_t^i), \quad (\text{B.1})$$

$$\begin{aligned}
\min_V \quad & F(V) = \frac{1}{2} \sum_{t=1}^T \sum_{i=1}^N \left\| \Gamma_{p_t \rightarrow \tilde{p}} \left(\text{Log}(p_t, y_t^i) - V_t x^i - g_t(z^i) \right) \right\|_F^2, \\
\text{s.t.} \quad & R(V) = \frac{1}{2} \sum_{t=1}^T \Lambda_t(V) \leq \lambda,
\end{aligned} \quad (\text{B.2})$$

where $\lambda > 0$ controls the level of smoothness constraint. Meanwhile, the specific form of the smoothing term $\Lambda_t(V)$ is as follows:

$$\Lambda_t(V) = \begin{cases} \|\gamma_2(V)\|_F^2, & t = 1, \\ \frac{1}{2}(\|\gamma_t(V)\|_F^2 + \|\gamma_{t+1}(V)\|_F^2), & 1 < t < T, \\ \|\gamma_T(V)\|_F^2, & t = T, \end{cases} \quad (\text{B.3})$$

Step 1: Updating V_t

Applying the logarithmic map to the regression model (B.1) projects the SPD-valued response variable onto the tangent space at p_t , leading to:

$$\text{Log}(p_t, y_t^i) = V_t x^i + g_t(z^i) + \varepsilon_t^i. \quad (\text{B.4})$$

Taking the conditional expectation with respect to z^i , we obtain:

$$\phi(z^i, p_t) = V_t \varphi(z^i) + g_t(z^i). \quad (\text{B.5})$$

Here we adopt a nonparametric kernel regression approach. Let $h > 0$ denote the bandwidth parameter. The kernel regression weights at location z are defined as $W_{j,h}(z) = \frac{K_h(z_j - z)}{\sum_{j'=1}^N K_h(z_{j'} - z)}$. Here, $K_h(\cdot)$ denotes a kernel function with bandwidth h , which is generally written as $K_h(u) = \frac{1}{h} K(\frac{u}{h})$, where $K(\cdot)$ is a symmetric, nonnegative kernel function satisfying $\int K(u) du = 1$. Hence, the nonparametric regression could be written as follow: $\hat{\phi}(z, p_t) = \sum_{j=1}^N W_{j,h_1}(z) \text{Log}(p_t, y_t^j)$, $\hat{\varphi}(z) = \sum_{j=1}^N W_{j,h_2}(z) x^j$. Subtracting equation (B.5) from (B.4) yields:

$$\text{Log}(p_t, y_t^i) - \phi(z^i, p_t) = V_t(x^i - \varphi(z^i)) + \varepsilon_t^i, \quad (\text{B.6})$$

Given that all matrix components reside in the tangent space $T_{p_t} \text{Sym}^+(d)$, the loss function for V_t is formulated as:

$$\begin{aligned} E(V) &= \frac{1}{2}F(V) + \frac{1}{2}R(V) \\ &= \frac{1}{2} \sum_{t=1}^T \sum_{i=1}^N \|\Gamma_{p_t \rightarrow \bar{p}}[\text{Log}(p_t, y_t^i) \\ &\quad - \hat{\phi}(z^i, p_t) - V_t(x^i - \hat{\varphi}(z^i))]\|_F^2 \\ &\quad + \frac{\lambda}{2} \sum_{t=2}^T \|\Gamma_{p_t \rightarrow \bar{p}} V_t - \Gamma_{p_{t-1} \rightarrow \bar{p}} V_{t-1}\|_F^2. \end{aligned} \quad (\text{B.7})$$

The gradient of $E(V)$ with respect to v_t^j is derived as:

$$\frac{\partial E}{\partial v_t^j} = \mathcal{D}_t^j + \lambda \begin{cases} -\delta_{t+1}^j, & t = 1, \\ \delta_t^j - \delta_{t+1}^j, & 1 < t < T, \\ \delta_t^j, & t = T, \end{cases} \quad (\text{B.8})$$

where $\mathcal{D}_t^j = -\sum_{i=1}^N \left(\Gamma_{p_t \rightarrow \tilde{p}} [\text{Log}(p_t, y_t^i) - \phi(z, p_t) - \sum_{j=1}^{d_x} v_t^j (x_j^i - \varphi_j(z))] \right) \cdot (x_j^i - \varphi_j(z))^T$ and $\delta_t^j = \Gamma_{p_t \rightarrow \tilde{p}} v_t^j - \Gamma_{p_{t-1} \rightarrow \tilde{p}} v_{t-1}^j$.

Step 2: Updating $g_t(z)$ By reintroducing the estimator $\widehat{V}_t$ back into equation (B.6), and simultaneously subtracting $\widehat{V}_t x^i$ from both sides of the equation, we obtain

$$\text{Log}(p_t, y_t^i) - \widehat{V}_t x^i = g_t(z^i) + \varepsilon_t^i. \quad (\text{B.9})$$

Similarly, employ kernel regression method to construct the estimator as

$$\widehat{g}_t(z) = \sum_{j=1}^N W_{j,h}(z) \left\{ \text{Log}(p_t, y_t^j) - \widehat{V}_t x^j \right\}. \quad (\text{B.10})$$

Where $W_{j,h}(z) = \frac{K_h(z_j - z)}{\sum_{j'=1}^N K_h(z_{j'} - z)}$.

Step 3: Updating p_t Finally, the base point p_t is optimized by minimizing:

$$\widehat{E}_N(p_t) = \frac{1}{2} \sum_{i=1}^N d_{\text{Sym}^+(d)}(\widehat{y}_t^i, y_t^i)^2 \quad (\text{B.11})$$

$$\begin{aligned} &= \frac{1}{2} \sum_{i=1}^N \|\text{Log}(\widehat{y}_t^i, y_t^i)\|_{\widehat{y}_t^i}^2 \\ &= \frac{1}{2} \sum_{i=1}^N \langle \text{Log}(\widehat{y}_t^i, y_t^i), \text{Log}(\widehat{y}_t^i, y_t^i) \rangle_{\widehat{y}_t^i} \\ &= \frac{1}{2} \sum_{i=1}^N \langle \text{Log}(\widehat{y}_t^i, y_t^i)_{\widehat{y}_t^i \rightarrow p_t}, \text{Log}(\widehat{y}_t^i, y_t^i)_{\widehat{y}_t^i \rightarrow p_t} \rangle_{p_t}, \end{aligned} \quad (\text{B.12})$$

where $\widehat{y}_t^i = \text{Exp}(\widehat{p}(t), \widehat{V}_t x^i + \widehat{g}_t(z^i))$, $\text{Log}(\widehat{y}_t^i, y_t^i)_{\widehat{y}_t^i \rightarrow p_t}$ represents the parallel transport of $\text{Log}(\widehat{y}_t^i, y_t^i)$ from $T_{\widehat{y}_t^i} \text{Sym}^+(d)$ to $T_{p_t} \text{Sym}^+(d)$. Here, the advantageous property of parallel transport in preserving inner products ensures

the validity of equations in (B.12). We then denote the gradient of $\widehat{E}_N(p_t)$ with respect to p_t as $\nabla_{p_t} \widehat{E}_N(p_t)$, following the main conclusions of [1], the gradient can be approximately expressed as

$$\nabla_{p_t} \widehat{E}_N(p_t) \approx - \sum_{i=1}^N \Gamma_{\hat{y}_t^i \rightarrow p_t} \text{Log}(\hat{y}_t^i, y_t^i). \quad (\text{B.13})$$

The corresponding gradient is:

$$\nabla_{p_t} \widehat{E}_N(p_t) \approx - \sum_{i=1}^N \Gamma_{\hat{y}_t^i \rightarrow p_t} \text{Log}(\hat{y}_t^i, y_t^i). \quad (\text{B.14})$$

The update step follows:

$$p_t = \text{Exp}(p_t, -\eta \nabla_{p_t} \widehat{E}_N(p_t)). \quad (\text{B.15})$$

Thus, by utilizing this approximate gradient as the direction for the gradient descent algorithm, iteratively optimizing $\widehat{E}_N(p_t)$ to achieve the estimation of the base point p_t .

Based on the above process, the estimation algorithm of the intrinsic semiparametric regression model can be summarized as Algorithm 1.

Appendix B.0.1. Algorithm Implementation

Algorithm 1 Estimation Procedure for Intrinsic Regression Model

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1: Input:  $x^i \in \mathbb{R}^{d_x}$ ,  $z^i \in \mathbb{R}^{d_z}$ ,  $y_t^i \in \text{Sym}^+(d)$ , where  $i = 1, \dots, N$ ,  $j = 1, \dots, T$ .
2: Output:  $p_t, V_t = \{v_t^j\}_{j=1}^{d_x}$  and  $g(t, z^i)$ .
3: Initialize  $p_t, V_t, a, a_{max}$ , centralize  $x^i, z^i$ , and construct estimator  $\hat{\varphi}(z)$  and estimator  $\phi(z^i, p_t)$ .
4: while termination condition do
5:   Calculate  $(v_t^j)_{\text{new}} = v_t^j - \eta \Gamma_{\tilde{p} \rightarrow p_t} \frac{\partial E}{\partial v_t^j}$ . according to (B.6);
6:   Calculate  $\hat{g}_t(z)$  according to (B.10);
7:   Calculate  $p_{t_{\text{new}}} = \text{Exp}(p_t, -a \nabla_{p_t} \hat{E}_N(p_t))$  according to (B.15);
8:   if  $E_N(p_{t_{\text{new}}}) < E_N(p_t)$  and  $E(V_{t_{\text{new}}}) < E(V_t)$  then
9:      $p_t \leftarrow p_{t_{\text{new}}}$ ;
10:     $V_t \leftarrow V_{t_{\text{new}}}$ ;
11:     $a = a_{max}$ ;
12:   else
13:      $a = a/2$ ;
14:   end if
15: end while

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References

- [1] Kim, H.J., Adluru, N., Collins, M.D., Chung, M.K., Bendlin, B.B., Johnson, S.C., Davidson, R.J., Singh, V., 2014. Multivariate general linear models (MGLM) on Riemannian manifolds with applications to statistical analysis of diffusion weighted images, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 2705–2712.
- [2] Wang, M., Wang, Y., Yang, Y., 2025. Dynamic and low-dimensional modeling of brain functional connectivity on riemannian manifolds. *NeuroImage*, 121243.